

analyze_summary.R

Juango the Blue

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```
# Setup -----

# Clear the environment and console
rm(list = ls())
cat("\014")

options(width = 1000)

# Import data
results <- read.csv("~/2016-Present (Michigan)/Research/12 Estimation Definitions/04 Analysis/Results/01 Results.csv")
names(results)[1:7] <- c('index', 'method', 'fn', 'probability', 'cycles', 'performance', 'degree')

# Create vectors of variables
fns = c("sphere", "ackley", "rosen", "stybtang")
methods = c("current", "future", "best")

# Organize Data -----

# Slice data by functions
sphere = results[results$fn == "sphere", c(1:2, 4:7)]
ackley = results[results$fn == "ackley", c(1:2, 4:7)]
rosen = results[results$fn == "rosenbrock", c(1:2, 4:7)]
stybtang = results[results$fn == "styblinski-tang", c(1:2, 4:7)]

# Copy over column names
names(sphere)[1:6] = names(results)[c(1:2, 4:7)]
names(ackley)[1:6] = names(results)[c(1:2, 4:7)]
names(rosen)[1:6] = names(results)[c(1:2, 4:7)]
names(stybtang)[1:6] = names(results)[c(1:2, 4:7)]

# Slice data by methods
ackley.current = ackley[ackley$method == "future_est" & ackley$probability == 0,]
ackley.future = ackley[ackley$method == "future_est" & ackley$probability == 1,]
ackley.best = ackley[ackley$method == "best_est" & ackley$probability == 1,]

# Slice data by probability
ackley.current.prob = ackley[ackley$method == "future_est" & ackley$performance > 0 & ackley$probability == 0,]
ackley.future.prob = ackley[ackley$method == "best_est" & ackley$performance > 0 & ackley$probability == 1,]

# Create scatterplot data
sphere.scatter = sphere[, c(4, 5)]
ackley.scatter = ackley[, c(4, 5)]
rosen.scatter = rosen[, c(4, 5)]
stybtang.scatter = stybtang[, c(4, 5)]

# Perform Analyses -----
```

Ackley T-Tests

```
ackley.tcyc.CF = t.test(ackley.current$cycles,ackley.future$cycles)
ackley.tcyc.CF
```

```
##
## Welch Two Sample t-test
##
## data:  ackley.current$cycles and ackley.future$cycles
## t = -0.91272, df = 170.43, p-value = 0.3627
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.790688  0.290688
## sample estimates:
## mean of x mean of y
##    15.60    15.85
```

```
ackley.tcyc.CB = t.test(ackley.current$cycles,ackley.best$cycles)
ackley.tcyc.CB
```

```
##
## Welch Two Sample t-test
##
## data:  ackley.current$cycles and ackley.best$cycles
## t = 0.77949, df = 186.78, p-value = 0.4367
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.2908532  0.6708532
## sample estimates:
## mean of x mean of y
##    15.60    15.41
```

```
ackley.tcyc.FB = t.test(ackley.future$cycles,ackley.best$cycles)
ackley.tcyc.FB
```

```
##
## Welch Two Sample t-test
##
## data:  ackley.future$cycles and ackley.best$cycles
## t = 1.47, df = 192.16, p-value = 0.1432
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.1503661  1.0303661
## sample estimates:
## mean of x mean of y
##    15.85    15.41
```

```
ackley.tperf.CF = t.test(ackley.current$performance,ackley.future$performance)
ackley.tperf.CF
```

```
##
## Welch Two Sample t-test
##
## data:  ackley.current$performance and ackley.future$performance
## t = -3.9207, df = 99, p-value = 0.0001628
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
```

```

## -27.407807 -8.988106
## sample estimates:
## mean of x mean of y
## 1.916607e-06 1.819796e+01

ackley.tperf.CB = t.test(ackley.current$performance,ackley.best$performance)
ackley.tperf.CB

##
## Welch Two Sample t-test
##
## data: ackley.current$performance and ackley.best$performance
## t = -3.4389, df = 99, p-value = 0.0008561
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -17.948134 -4.814301
## sample estimates:
## mean of x mean of y
## 1.916607e-06 1.138122e+01

ackley.tperf.FB = t.test(ackley.future$performance,ackley.best$performance)
ackley.tperf.FB

##
## Welch Two Sample t-test
##
## data: ackley.future$performance and ackley.best$performance
## t = 1.1958, df = 178.99, p-value = 0.2334
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -4.432372 18.065851
## sample estimates:
## mean of x mean of y
## 18.19796 11.38122

# Run regression on intermediate data (probability = 0.1 to 0.9)
ackley.rcyc.C = lm(ackley.current.prob$cycles ~ ackley.current.prob$probability)
summary(ackley.rcyc.C)

##
## Call:
## lm(formula = ackley.current.prob$cycles ~ ackley.current.prob$probability)
##
## Residuals:
## Min 1Q Median 3Q Max
## -3.5287 -1.5207 -0.5207 0.4793 9.4740
##
## Coefficients:
## Estimate Std. Error t value Pr(>|t|)
## (Intercept) 15.54200 0.09938 156.395 <2e-16 ***
## ackley.current.prob$probability -0.02667 0.18615 -0.143 0.886
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.691 on 998 degrees of freedom
## Multiple R-squared: 2.056e-05, Adjusted R-squared: -0.0009814

```

```
## F-statistic: 0.02052 on 1 and 998 DF, p-value: 0.8861
ackley.rperf.C = lm(ackley.current.prob$performance ~ ackley.current.prob$probability)
summary(ackley.rperf.C)

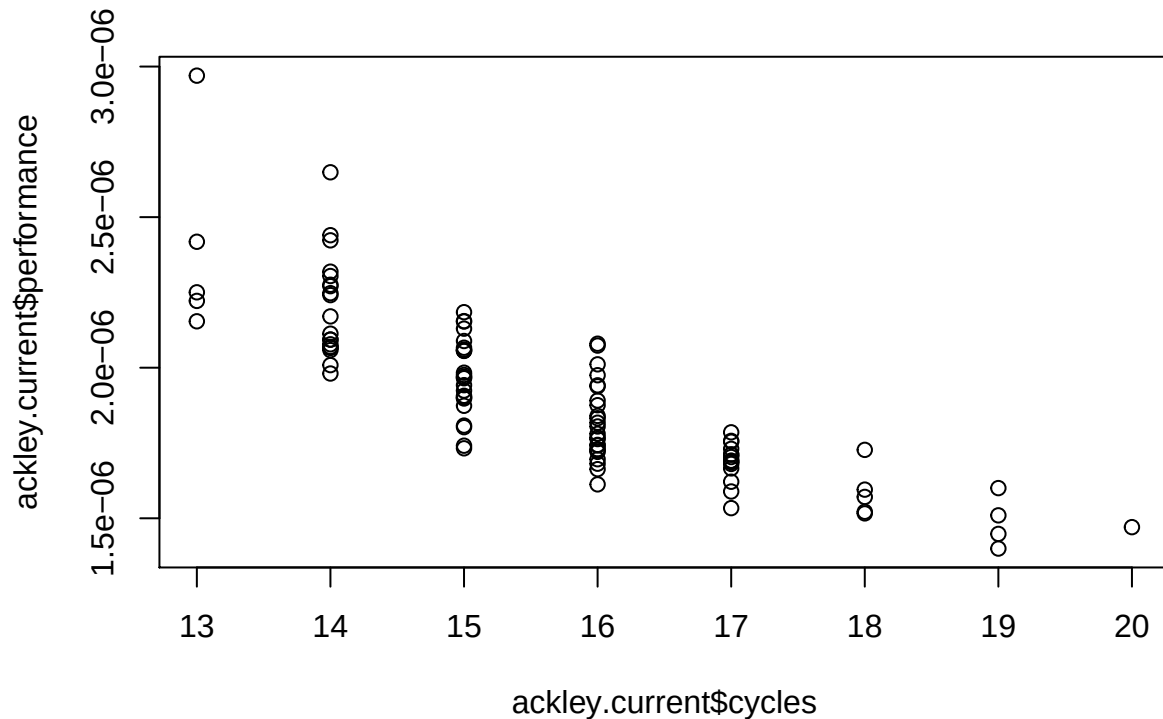
##
## Call:
## lm(formula = ackley.current.prob$performance ~ ackley.current.prob$probability)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.492  -3.943  -1.620  -0.071  174.745
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.4779     0.7493  -1.972   0.0488 *
## ackley.current.prob$probability  7.7448     1.4036   5.518 4.37e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.75 on 998 degrees of freedom
## Multiple R-squared:  0.02961, Adjusted R-squared:  0.02863
## F-statistic: 30.45 on 1 and 998 DF, p-value: 4.372e-08
ackley.rcyc.F = lm(ackley.future.prob$cycles ~ ackley.future.prob$probability)
summary(ackley.rcyc.F)

##
## Call:
## lm(formula = ackley.future.prob$cycles ~ ackley.future.prob$probability)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.5845 -1.4289 -0.4484  0.7968  6.5711
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    15.4094     0.0969 159.017 <2e-16 ***
## ackley.future.prob$probability  0.1946     0.1815   1.072   0.284
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.649 on 998 degrees of freedom
## Multiple R-squared:  0.00115, Adjusted R-squared:  0.0001488
## F-statistic: 1.149 on 1 and 998 DF, p-value: 0.2841
ackley.rperf.F = lm(ackley.future.prob$performance ~ ackley.future.prob$probability)
summary(ackley.rperf.F)

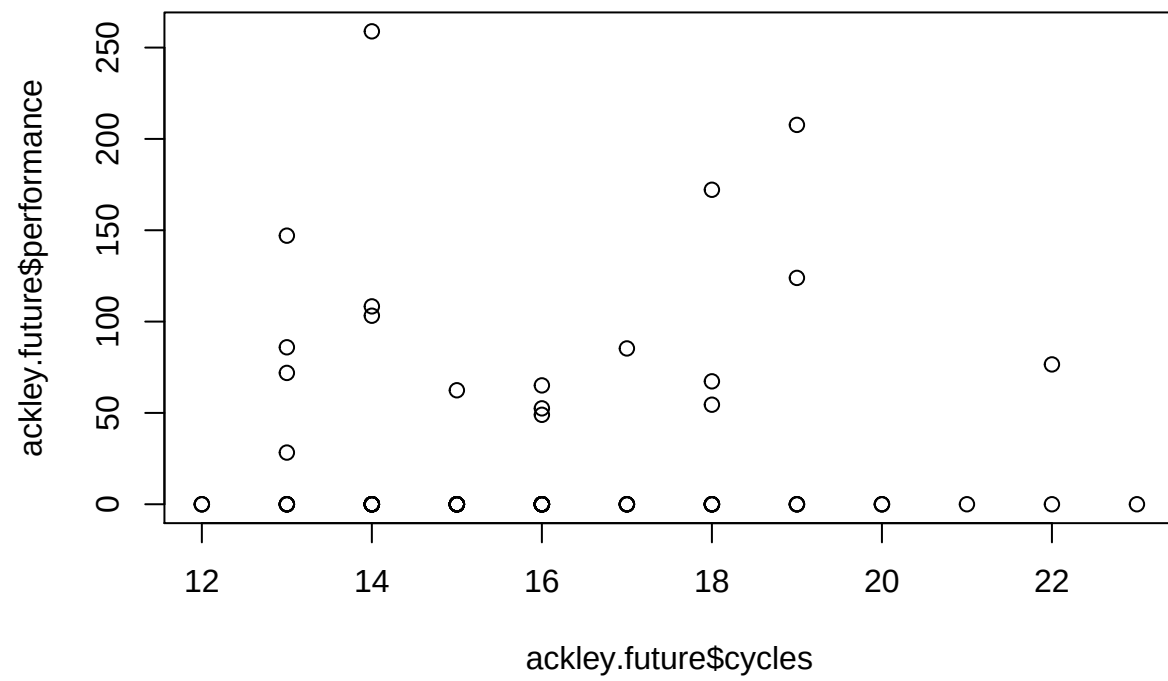
##
## Call:
## lm(formula = ackley.future.prob$performance ~ ackley.future.prob$probability)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.37  -6.55  -2.33   0.49  325.87
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -3.301      1.366  -2.416   0.0159 *
## ackley.future.prob$probability  14.073      2.559    5.499 4.85e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23.25 on 998 degrees of freedom
## Multiple R-squared:  0.02941,    Adjusted R-squared:  0.02844
## F-statistic: 30.24 on 1 and 998 DF,  p-value: 4.847e-08
```

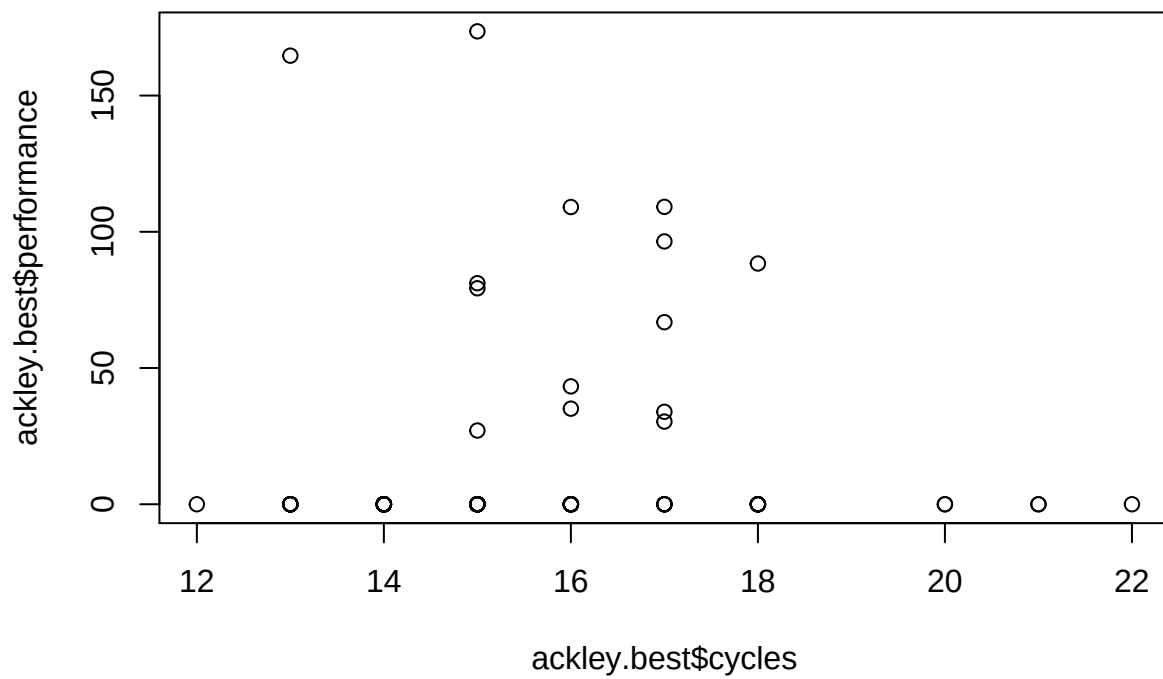
```
# Plot Data -----
# Generate scatterplots
plot(ackley.current$cycles,ackley.current$performance)
```



```
plot(ackley.future$cycles,ackley.future$performance)
```

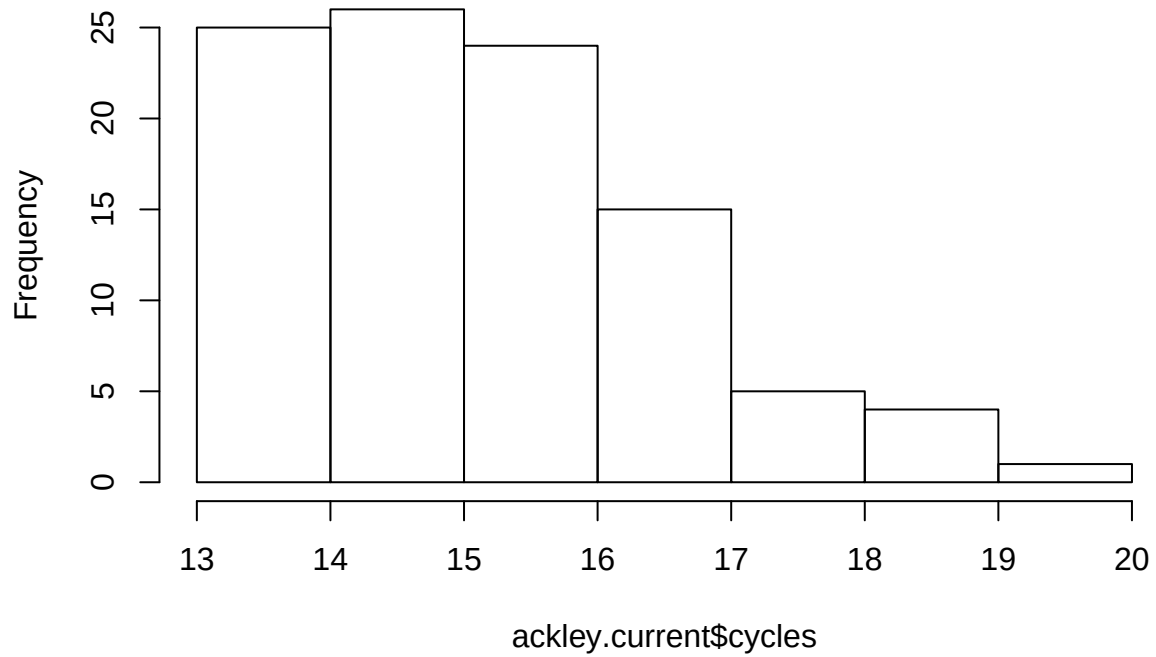


```
plot(ackley.best$cycles,ackley.best$performance)
```



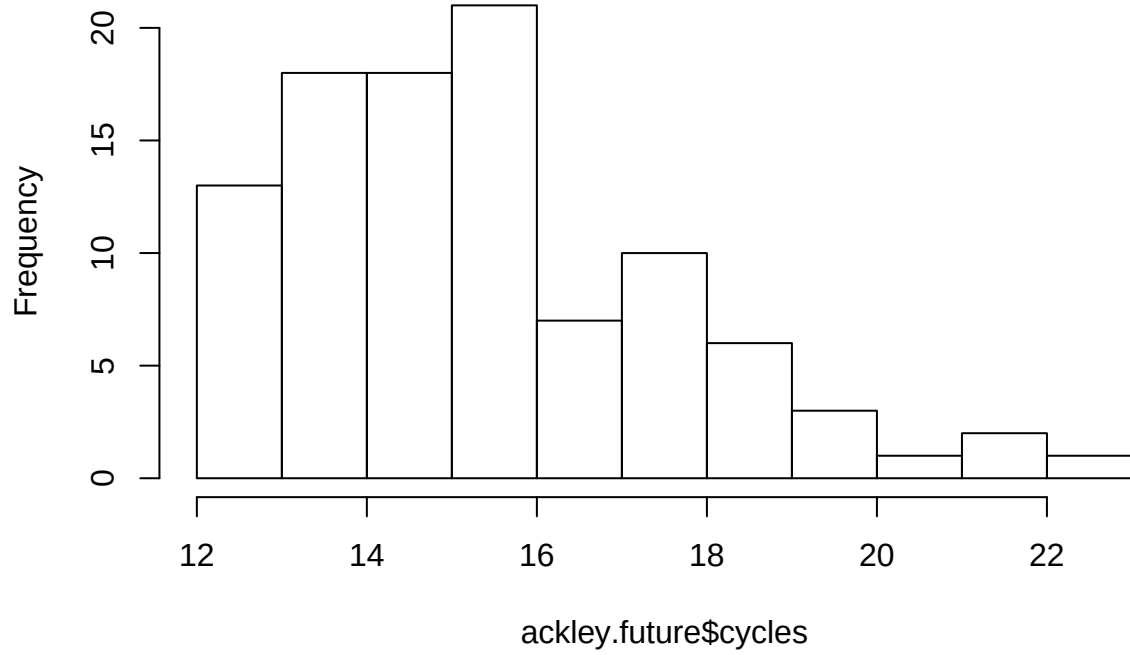
```
# Generate ackley histograms  
hist(ackley.current$cycles)
```

Histogram of ackley.current\$cycles



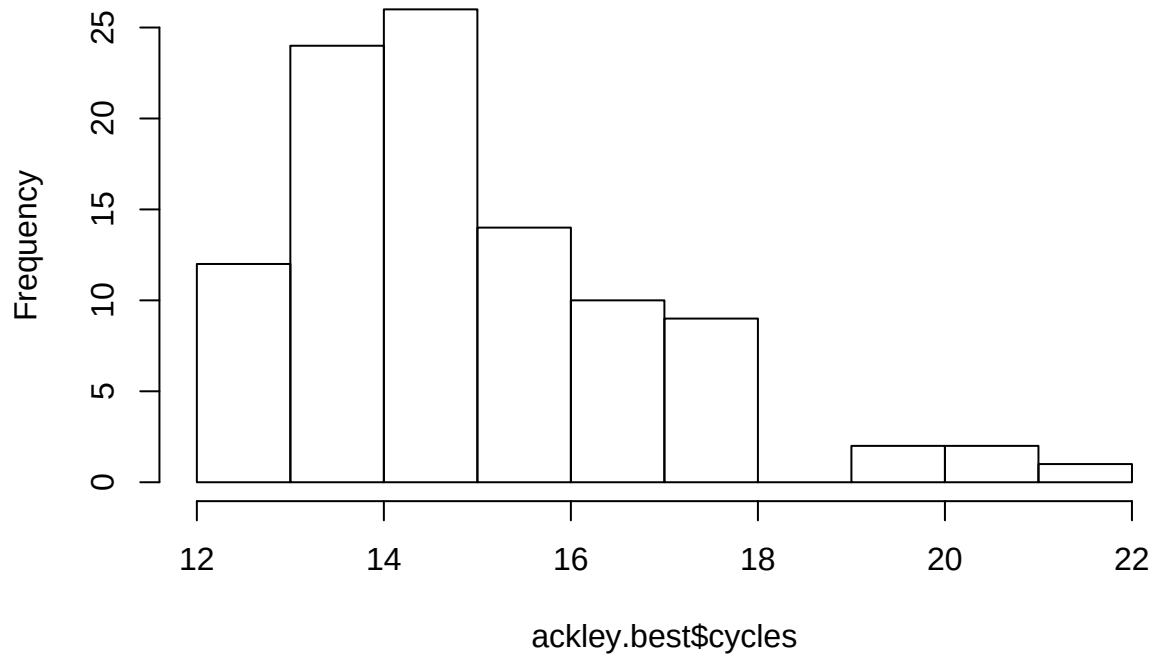
```
hist(ackley.future$cycles)
```


Histogram of ackley.future\$cycles



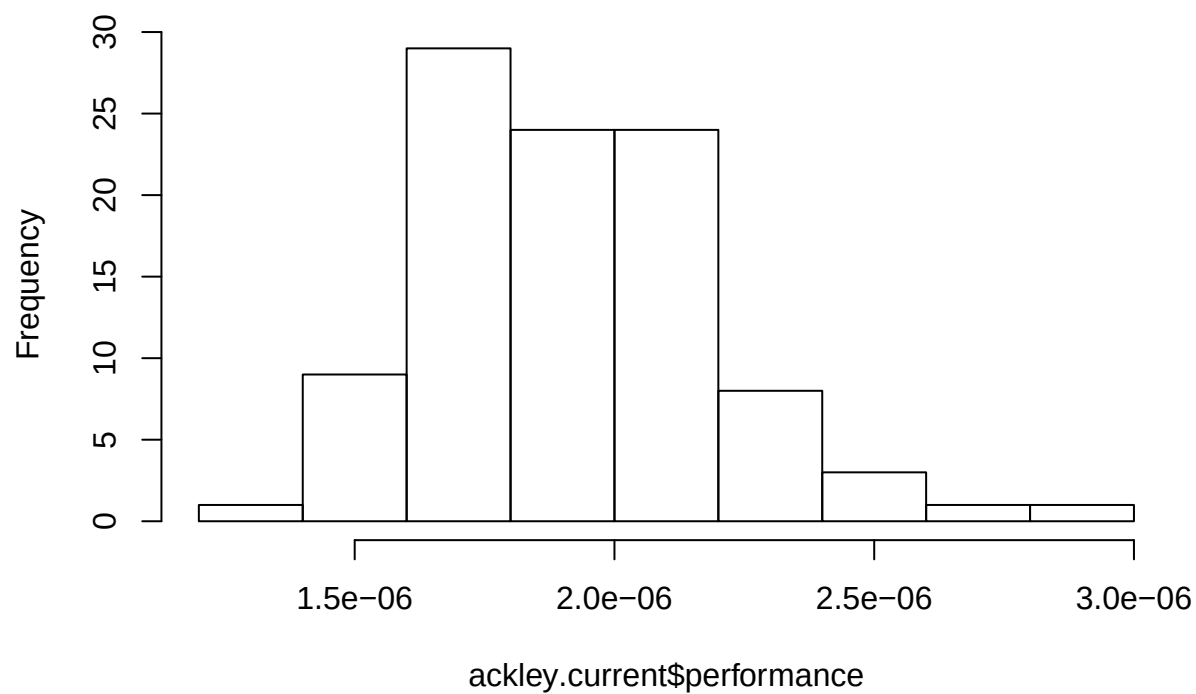
```
hist(ackley.best$cycles)
```

Histogram of ackley.best\$cycles



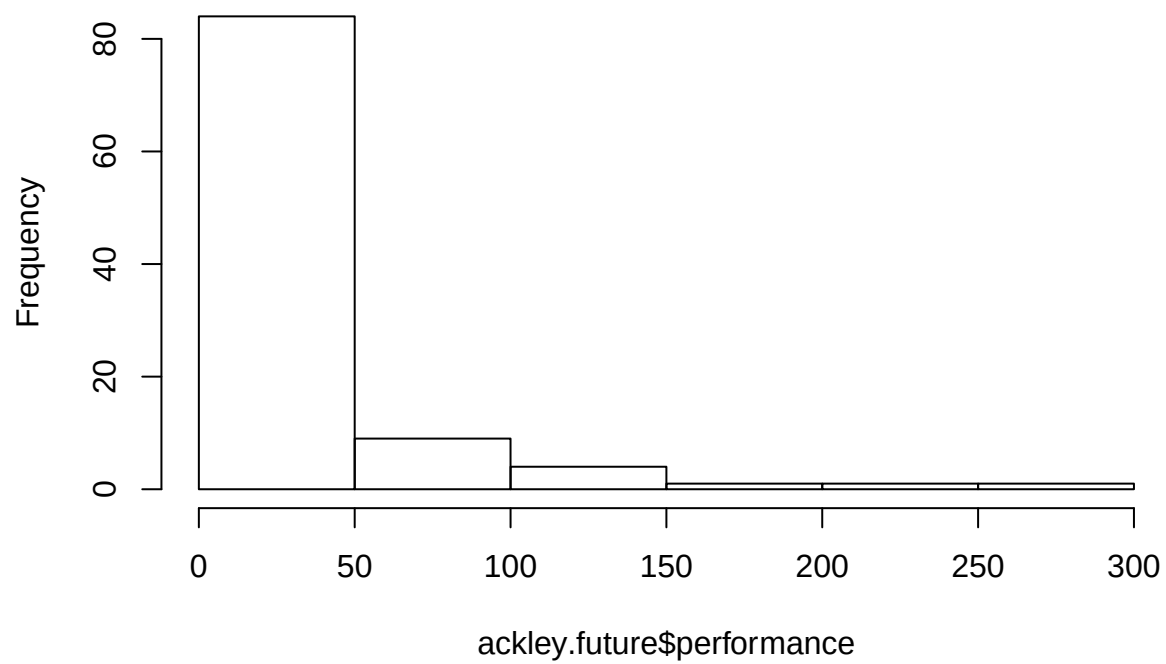
```
hist(ackley.current$performance)
```

Histogram of ackley.current\$performance



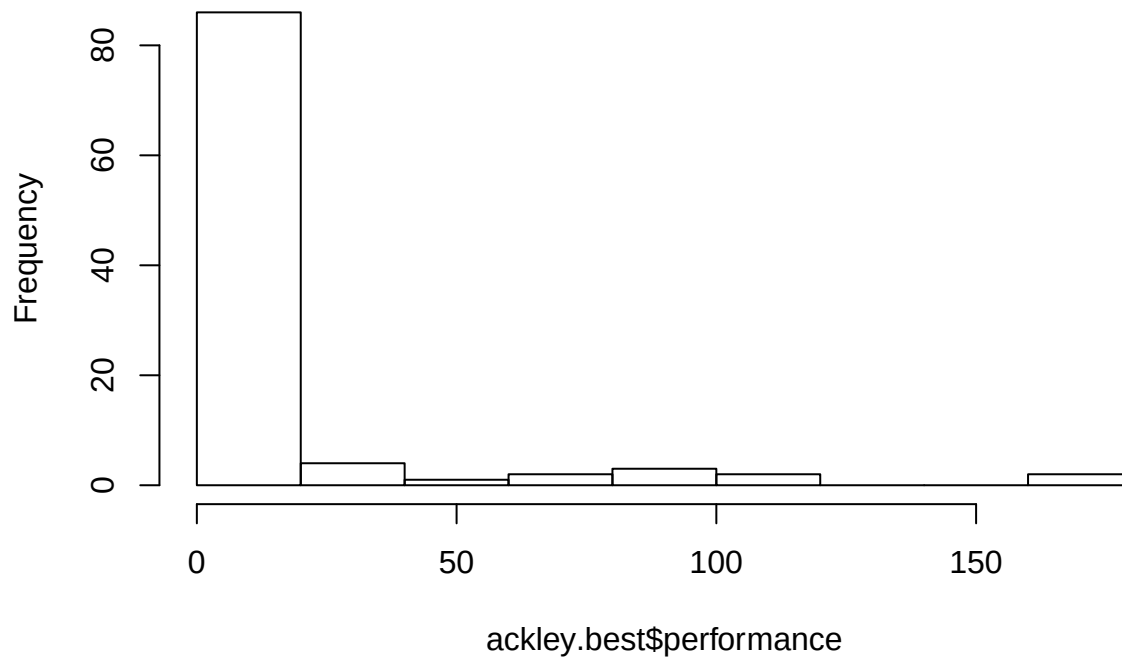
```
hist(ackley.future$performance)
```

Histogram of ackley.future\$performance



```
hist(ackley.best$performance)
```

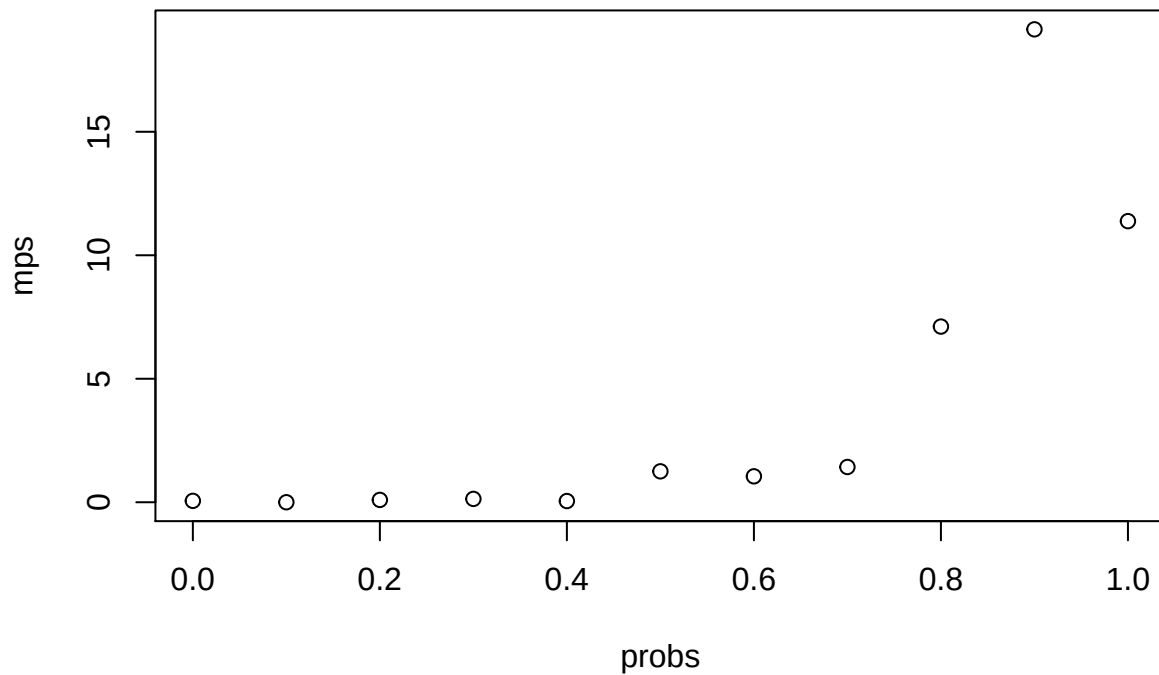
Histogram of ackley.best\$performance



```
# Plot mean performances
```

```
mp00 = mean(ackley$performance[ackley$probability == 0 & ackley$method == "best_est"])
mp01 = mean(ackley$performance[ackley$probability == 0.1 & ackley$method == "best_est"])
mp02 = mean(ackley$performance[ackley$probability == 0.2 & ackley$method == "best_est"])
mp03 = mean(ackley$performance[ackley$probability == 0.3 & ackley$method == "best_est"])
mp04 = mean(ackley$performance[ackley$probability == 0.4 & ackley$method == "best_est"])
mp05 = mean(ackley$performance[ackley$probability == 0.5 & ackley$method == "best_est"])
mp06 = mean(ackley$performance[ackley$probability == 0.6 & ackley$method == "best_est"])
mp07 = mean(ackley$performance[ackley$probability == 0.7 & ackley$method == "best_est"])
mp08 = mean(ackley$performance[ackley$probability == 0.8 & ackley$method == "best_est"])
mp09 = mean(ackley$performance[ackley$probability == 0.9 & ackley$method == "best_est"])
mp10 = mean(ackley$performance[ackley$probability == 1.0 & ackley$method == "best_est"])
```

```
probs = c(0,0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9,1.0)
mps = c(mp00,mp01,mp02,mp03,mp04,mp05,mp06,mp07,mp08,mp09,mp10)
plot(probs,mps)
```



```
# Plot mean cycles
```

```
mc00 = mean(ackley$cycles[ackley$probability == 0 & ackley$method == "best_est"])
mc01 = mean(ackley$cycles[ackley$probability == 0.1 & ackley$method == "best_est"])
mc02 = mean(ackley$cycles[ackley$probability == 0.2 & ackley$method == "best_est"])
mc03 = mean(ackley$cycles[ackley$probability == 0.3 & ackley$method == "best_est"])
mc04 = mean(ackley$cycles[ackley$probability == 0.4 & ackley$method == "best_est"])
mc05 = mean(ackley$cycles[ackley$probability == 0.5 & ackley$method == "best_est"])
mc06 = mean(ackley$cycles[ackley$probability == 0.6 & ackley$method == "best_est"])
mc07 = mean(ackley$cycles[ackley$probability == 0.7 & ackley$method == "best_est"])
mc08 = mean(ackley$cycles[ackley$probability == 0.8 & ackley$method == "best_est"])
mc09 = mean(ackley$cycles[ackley$probability == 0.9 & ackley$method == "best_est"])
mc10 = mean(ackley$cycles[ackley$probability == 1.0 & ackley$method == "best_est"])
```

```
mcs = c(mc00,mc01,mc02,mc03,mc04,mc05,mc06,mc07,mc08,mc09,mc10)
plot(probs,mcs)
```

